

Er³⁺-doped tellurite glasses for enhancing a solar cell photocurrent through photon upconversion upon 1500 nm excitation

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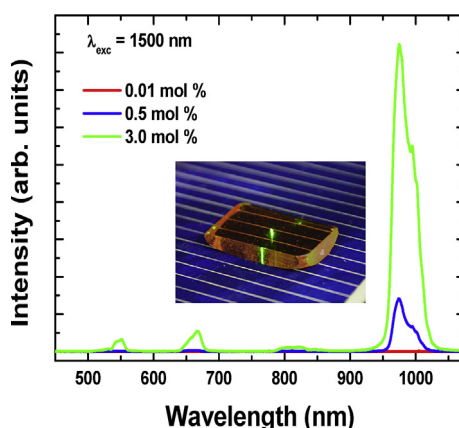
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HIGHLIGHTS

- Er³⁺-doped tungstate-tellurite glasses were synthesized by melt-quenching technique.
- Bright upconversion emission at 980 nm was achieved by excitation with 1500 nm laser.
- Solar cell photo-current was enhanced in the presence of 3.0 mol% Er³⁺-doped glass.
- Upconversion process has been confirmed by the temporal evaluation studies.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 14 July 2016

Received in revised form

28 April 2017

Accepted 3 June 2017

Available online 3 June 2017

Keywords:

Solar-cell

Upconversion

Lanthanide doped tellurite glasses

ABSTRACT

Trivalent erbium-doped tellurite glasses have been prepared and investigated with an aim to improve the efficiency of a Silicon (Si) solar cell through photon upconversion processes. The upconversion emissions at 548 nm ($^4S_{3/2}$, $^2H_{11/2} \rightarrow ^4I_{15/2}$), 671 nm ($^4F_{9/2} \rightarrow ^4I_{15/2}$), 800 nm ($^4I_{9/2} \rightarrow ^4I_{15/2}$) and 975 nm ($^4I_{11/2} \rightarrow ^4I_{15/2}$) of Er³⁺ ions have been observed under 1500 nm laser excitation. The 3.0 mol% Er₂O₃-doped glass exhibits high upconversion emission at 975 nm compared to 0.01 and 0.5 mol% Er₂O₃-doped glasses. Temporal evolution studies confirm that energy transfer upconversion is responsible for observed upconversion emissions, which is in good agreement with a simple rate-equations model. The upconversion intensity and photocurrent of solar cell are found to increase with the increase of pump power of the laser. A significant enhancement in photocurrent of a solar cell has been observed, when the

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Efficiency
Rate equations

3.0 mol% Er_2O_3 -doped glass is placed over a solar cell under 1500 nm laser excitation. All the results indicate that 3.0 mol% Er_2O_3 -doped glass could be suitable for enhancing the solar cell efficiency.

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1. Introduction

The researchers are striving for the development of advanced materials for solar energy harvesting to meet ever increasing demand of domestic and industrial requirements with an aim to reduce environmental pollution [1]. In spite of significant developments in the photovoltaic industry, the efficient and cost-effective conversion of solar energy into electricity through solar cells remains a daunting task. The conversion efficiency of solar cells is not very efficient due to thermalization of high energy photons (<500 nm) and transmission of low energy photons (>1150 nm) of solar spectrum through solar cell. These problems can be overcome by modifying the solar spectrum through down-conversion (quantum cutting) and up-conversion mechanisms. The thermalization losses can be reduced by the conversion of a high energy photon to two or more low energy photons whose energy slightly greater or equal to energy gap of solar cell through down conversion (DC) process [2,3]. On the other hand, transmission losses can be reduced by converting low energy infrared photons to high energy near infrared or visible photons whose energy slightly greater or equal to the bandgap energy of Si solar cell through photon upconversion (UC) processes [4–8].

Therefore, up and down conversion materials such as Ln^{3+} ions doped glasses, glass-ceramics, and phosphors modify the incident solar spectrum to enhance the efficiency of the solar cell. The DC materials convert a high energy photon to two or more low energy near infrared photons [3] whereas UC materials convert low energy infrared photons to high energy near infrared or visible photons [4] whose energies slightly greater or equal to the energy gap of Si solar cell. Hence, the DC or UC converted photons are absorbed by the solar cell and creates an additional electron–hole pairs to generate photocurrent for improving the solar cell efficiency [9]. The UC and DC processes are utilized not only for harvesting of solar energy but also for enhancing the photocatalytic activity [1].

Among the Ln^{3+} ions, Er^{3+} ions are widely used for enhancing the efficiency of c-Si solar cells since they can convert long wavelength 1550 nm photons into short wavelength photons through excited state absorption and/or energy transfer upconversion. When Er^{3+} ions are excited by photons with wavelength of about 1550 nm it produces four upconversion emission bands; $^4\text{I}_{11/2} \rightarrow ^4\text{I}_{15/2}$ (980 nm), $^4\text{I}_{9/2} \rightarrow ^4\text{I}_{15/2}$ (810 nm), $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ (660 nm) and $^4\text{S}_{3/2}, ^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$ (550 nm) [10]. These emission bands are well matched with the absorption region of c-Si solar cell. The Er^{3+} ion absorb the solar radiation around 1.54 μm and gives the emission in the visible and near infrared regions. The energy of these emissions is well above the Si band gap, absorbed by the solar cell creates additional electron–hole pairs to generate photocurrent in it [9] for harvesting of solar energy.

Among the oxide based glasses, tellurite glasses have the advantages of comparatively large Ln^{3+} ions solubility, high third order non-linearity, high transparency in the ultra-violet and near infra-red regions, low vibrational frequency compared to other oxide glasses and high refractive index that creates large absorption and emission cross-sections [11]. Hence, the tellurite glasses have been employed for the fabrication of optical fibers and planar waveguides [12]. The tellurite glasses of chemical composition $\text{TeO}_2\text{-WO}_3\text{-ZrO}_2$ have been chosen as a host for Er^{3+} ions in the

present investigation since TeO_2 is a good glass former and it increases the glass forming ability with the addition of heavy metal and transition metal oxides. Glass modifier, ZrO_2 increases the viscosity of the glass transition and melting temperatures, whereas WO_3 increases the resistance of devitrification and chemical stability.

Ln^{3+} -doped materials have been investigated for enhancing the efficiency of the solar cell. Gibart et al. [13] first reported the application of upconversion phosphors for improving the performance of solar cells in 1996. They combined the GaAs solar cell with a vitroceraic material doped with Yb^{3+} and Er^{3+} ion and enhanced the solar cell efficiency by 2.5% under high excitation densities [13]. Since then many researchers have been involved in the development of upconverting materials for enhancing the efficiency of solar cells. Recently, enhancement of photocurrent by utilizing UC process has been reported in $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped fluorindate [5] and oxyhalide-tellurite [6] glasses upon excitation with 975 and 1480 nm, respectively. Acoustic and optical properties of $\text{TeO}_2\text{-WO}_3\text{-ZrO}_2$ glasses were also reported elsewhere [14]. Spectroscopic properties of Sm^{3+} , Eu^{3+} and Er^{3+} -doped $\text{TeO}_2\text{-WO}_3\text{-ZrO}_2$ glasses were investigated [15–17]. So far, there have been no reports on Er^{3+} -doped $\text{TeO}_2\text{-WO}_3\text{-ZrO}_2$ glasses for improving the photocurrent of solar cell by UC processes under 1500 nm excitation. Hence, the present work aims to prepare Er^{3+} -doped tellurite glasses and study concentration dependent UC properties upon 1500 nm excitation for improving photocurrent of solar cell. The photocurrent of the solar cell was tested by placing the tellurite glass over a Si solar cell and then its performance was discussed in detail by considering a rate-equations model for explaining the ETU process.

2. Experimental details

Er^{3+} -doped tellurite glasses of molar composition, $74\text{TeO}_2\text{-(18-x)-WO}_3\text{-8ZrO}_2\text{-xEr}_2\text{O}_3$ ($x = 0.01, 0.5$ and 3.0 mol %) were prepared by conventional melt-quenching technique from high purity (99.9%, Aldrich) reagents (TeO_2 , WO_3 , ZrO_2 and Er_2O_3). The selected composition was melted in a platinum crucible for 40 min at 850 °C, followed by quenching glass melt in air on a brass mold that was pre-heated at glass-transition temperature (T_g). The samples were annealed at 280 °C (T_g) for 10 h to suppress thermal stress/strains and cool down to room temperature. Finally, they were cut and polished at uniform thickness for the spectroscopic measurements.

The optical absorption spectra were measured using a spectrophotometer (Perkin Elmer Lambda–950). The upconversion spectra were obtained by exciting with a commercial 1500 nm laser with a maximum power of 80 mW. A beam of laser was focused on the samples using a lens with a focal length of 30 mm. The emitted light was collected with a lens of 50 mm of focal length and the emission was focused into the entrance slit of a CCD spectrograph (Andor Shamrock 303i). The luminescence decay curves were obtained by exciting with 10 ns pulsed optical parametric oscillator laser (EKSPLA/NT342/3/UVE) and the emission was focused into the entrance of the spectrograph which was coupled to a photomultiplier (R928 Hamamatsu in the visible range or 5509-73 Hamamatsu for the near infrared (NIR) range) and using a digital storage oscilloscope (Tektronix 2430).

3. Results and discussion

3.1. Optical absorption

The absorption spectra for different concentrations of Er_2O_3 -doped tellurite glasses in the visible and near infra-red regions are shown in Fig. 1. The inset of Fig. 1 shows a close inspection of the absorption region between 920 and 1040 nm with a peak absorption at 975 nm which is a characteristic band of Er^{3+} ions [18]. The absorption spectrum shows that there is no sign of Yb^{3+} absorption which appears usually in the 900–1100 nm region [19]. In addition, the absorbance of the 0.01 mol% Er_2O_3 -doped glass is higher than the 0.5 mol% Er_2O_3 -doped glass due to surface imperfections, which may lead to scattering loss. The spectra consist of eight bands correspond to intra-configurational $4f^{11}-4f^{11}$ electronic transitions starting from the $^4I_{15/2}$ ground state to the different excited levels of the Er^{3+} ion.

3.2. Upconversion

The upconversion is a phenomenon of absorption of two or more low energy photons via intermediate long lived energy states followed by the emission of high energy photons. Fig. 2 shows the UC emission spectra of tellurite glasses doped with different Er_2O_3 concentrations upon 1500 nm excitation. The characteristic emission bands of Er^{3+} ions centered at around 548, 671, 800 and 975 nm corresponds to the ($^4S_{3/2}, ^2H_{11/2}$) $\rightarrow$ $^4I_{15/2}$, $^4F_{9/2} \rightarrow ^4I_{15/2}$, $^4I_{9/2} \rightarrow ^4I_{15/2}$ and $^4I_{11/2} \rightarrow ^4I_{15/2}$ transitions, respectively have been observed. Fig. 3 represents the partial energy level structure of Er^{3+} ion in tellurite glasses along with observed emission channels. As can be seen from Fig. 2, it is interesting to note that the UC emission enhances with the increase of Er_2O_3 concentration (up to 3.0 mol %). Moreover, the emission band at 975 nm is more intense compared to all other emission bands, which matches well with the band gap energy of c-Si solar cell. The reason for this intense emission and its dependence on photocurrent of solar cell is explained in the subsequent sections.

Pump power dependence studies were carried out for investigating the number of photons that are involved in the UC process. The dependence of upconversion intensity with the pump power follows the expression [20], $I = AC^P$, where I is the intensity of UC emission, A is a constant, C is the power of the pumping laser and P represents the number of photons responsible for UC process. Fig. 4 shows the UC emission band intensity at 975 nm of Er^{3+} ions in tellurite glasses upon 1500 nm laser excitation as a function of pump power. The value of P is found to be 1.8 for 975 nm emission of Er^{3+} ion. It indicates that two 1500 nm photons are responsible for the UC emission at 975 nm. Moreover, the dependence of UC emission shows a quadratic behavior. This behavior increases significantly with the increase of excitation power of laser at 1500 nm. These results evidences that two near infra-red photons are essential to populate the $^4I_{11/2}$ level for UC process.

The most efficient mechanisms for upconversion process are two-step excitation (GSA followed by excited state absorption (ESA)) and energy transfer upconversion (ETU). In ESA, successive absorption of pump photons in a single ion occurs and it depends on excitation power. But in the ETU process, sequential absorption of two photons by two neighbouring Er^{3+} ions are involved to populate their metastable (intermediate) levels. Then fluorescent level of one of the active ions has been populated through energy transfer between them. The ETU can be distinguished from ESA by comparing luminescent decay and dynamics curves. Although, the ESA process takes place in single Ln^{3+} ions (e.g., Er^{3+}), the weak absorption induced by parity-forbidden intra-4f transitions within Ln^{3+} strongly suppresses its efficiency. By contrast, the ETU takes

advantage of the high absorption cross-section and thus it is established as the most efficient UC process in practical applications. Furthermore, the energy transfer processes also depend on the multipolar interaction (dipole-dipole, dipole-quadrupole, quadrupole-quadrupole) between the donor and acceptor of Er^{3+} ions [21,22].

In order to know the dominant upconversion mechanism i.e., whether it is due to ESA or ETU or both, the temporal evolution of upconversion luminescence was measured under 1500 nm laser excitation. The temporal evolution curve for the $^4I_{11/2} \rightarrow ^4I_{15/2}$ transition (975 nm) was measured under 1500 nm pulsed laser excitation, as shown in Fig. 5. These curves show a rise time which indicates the Er^{3+} ion transferring its energy to a nearby Er^{3+} ion, and is promoted to higher excited states (see Fig. 3). It is confirmed that the ETU process is considered to be the most efficient in the Er^{3+} -doped tellurite glasses as evidenced by the temporal evolution of the UC emission. Phonon energy of the glass plays a crucial role in the multiphonon relaxation rate. The maximum phonon energy of the glass system is 920 cm^{-1} [17]. A gap with an energy of 3800 cm^{-1} (between $^4I_{11/2}$ and $^4I_{13/2}$ levels) requires about 4–5 phonons to bridge. The multiphonon relaxation rate can be calculated by the following equation [23]: $W_{MP} = \frac{1}{\tau_{exp}} - \frac{1}{\tau_{rad}}$, where τ_{exp} (1.85 ms) and τ_{rad} (3.85 ms) are the experimental and radiative lifetime of $^4I_{11/2}$ level. The τ_{rad} is obtained from the Judd-Ofelt analysis which is carried out for 0.5 mol% Er_2O_3 -doped glass. The multiphonon relaxation rate is found to be 280 s^{-1} , which is responsible for population of $^4I_{11/2}$ level.

Decay curves for the $^4I_{13/2}$ level were also measured under 1500 nm excitation are shown in Fig. 6. Decay curves exhibit a single exponential behavior with lifetime of 2.45, 2.00 and 0.72 ms for the 0.01, 0.5 and 3.0 mol% of Er_2O_3 -doped glasses, respectively. The quenching of lifetime with increasing Er^{3+} concentration is due to energy transfer process. The $^4I_{13/2}$ level has long fluorescence lifetime and hence acts as a reservoir for excited Er^{3+} ions. Moreover, $^4I_{13/2}$ level has a longer lifetime compared to all other fluorescent levels that were participated in the UC process.

Decay curves for the $^4I_{11/2}$ level were also measured under 800 nm excitation by monitoring emission at 975 nm, and they found nearly exponential nature. The average lifetime decreases

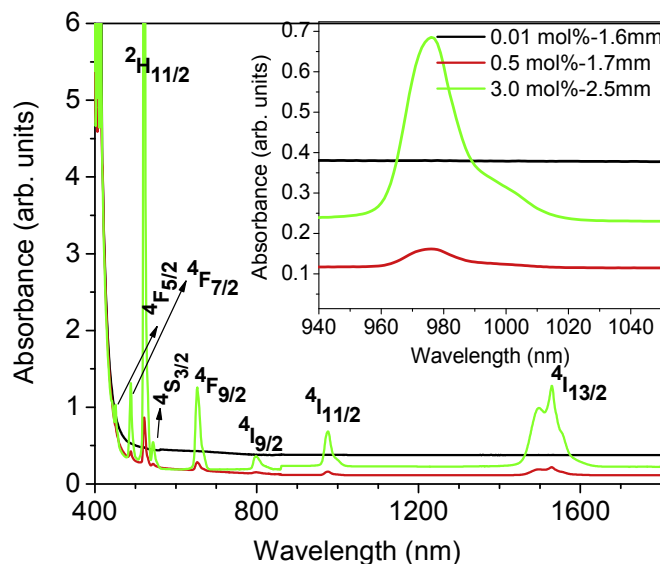


Fig. 1. Absorption spectra of Er^{3+} -doped tellurite glasses as a function of Er_2O_3 concentration in the visible and near infra-red regions. Inset shows a close inspection of the absorption spectra in the range of 920–1040 nm.

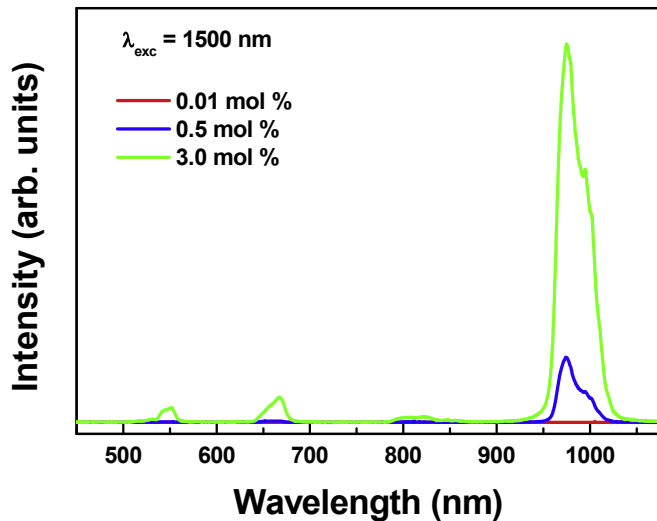


Fig. 2. Upconversion emission spectra of Er^{3+} -doped tellurite glasses as a function of Er_2O_3 concentration upon laser excitation at 1500 nm.

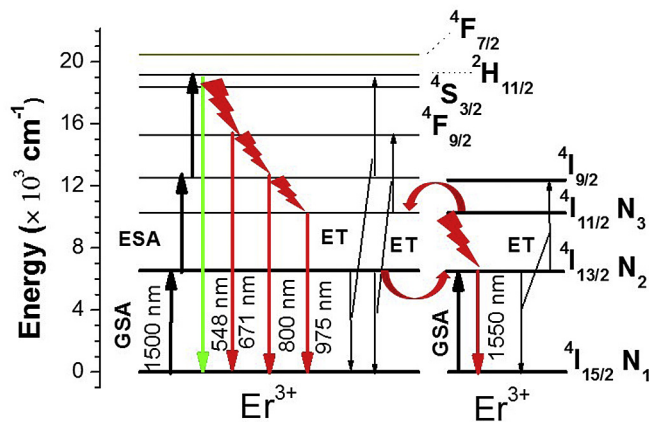


Fig. 3. Energy level diagram of Er^{3+} ion, emission transitions and upconversion channels under laser excitation at 1500 nm.

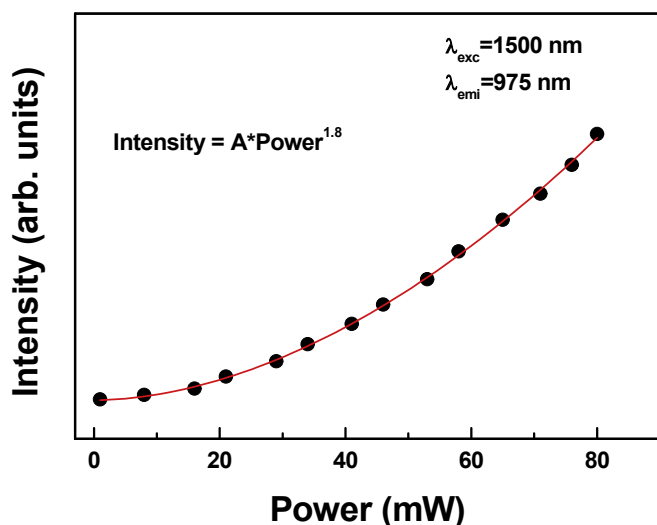
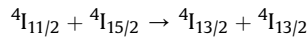


Fig. 4. Upconversion intensity of 975 nm emission band for 3.0 mol% Er_2O_3 -doped glass as a function of pump power with a laser at 1500 nm.

with the increase of the Er^{3+} concentration. The average lifetime is found to be 69, 57 and 42 μs for the glasses doped with 0.01, 0.5 and 3.0 mol% Er_2O_3 , respectively. These results indicate that the cross-relaxation process depopulates the $^4\text{I}_{11/2}$ level according to the following non-resonant channel:



A rate equations model is proposed to analyse the UC processes. According to this model, depending on the population at N_2 and N_3 levels, the rate equations are written as follows:

$$\frac{dN_2}{dt} = \sigma\phi N_1 - \frac{1}{\tau_2} N_2 - W_{UP} N_2^2 \quad (1)$$

$$\frac{dN_3}{dt} = -\frac{1}{\tau_3} N_3 + W_{UP} N_2^2 \quad (2)$$

where N_i corresponds to the population of the i^{th} level of Er^{3+} ions (see Fig. 3), σ is the absorption cross section of Er^{3+} ion at 1500 nm with an incident pump flux, ϕ , W_{UP} is the UC energy transfer rate and τ_i is the lifetime of i^{th} level of Er^{3+} ion.

The population of excited levels as well as the efficiency of the UC process is relatively low. The UC efficiencies have been reported to be about 3% for the $\beta\text{-NaYF}_4:\text{Yb}^{3+}, \text{Er}^{3+}$ sample under excitation with 943 nm laser and a laser power of 20 W/cm^2 [24]. Therefore, neglecting the ground state depopulation (N_1 is approximately the concentration of Er^{3+} ions) and the transfer term in Eq. (1) upon intrinsic processes (non-radiative and multiphonon relaxations). The number of Er^{3+} ion population at $^4\text{I}_{11/2}$ level for a steady state condition is written as:

$$N_3 = \tau_3 \tau_2^2 W_{UP} (\sigma\phi)^2 N_1^2 \quad (3)$$

It is emphasised that Eq. (3) predicts a quadratic behavior of the population of excited Er^{3+} ions at the N_3 level ($^4\text{I}_{11/2}$) which depends on the pump intensity, ϕ . This dependency is in agreement with the experimental results that are shown in Fig. 4. The value of $P(1.8)$ is close to the predicted value of 2.0 according to Eq. (3), which indicates two photons are responsible for the observed UC emission.

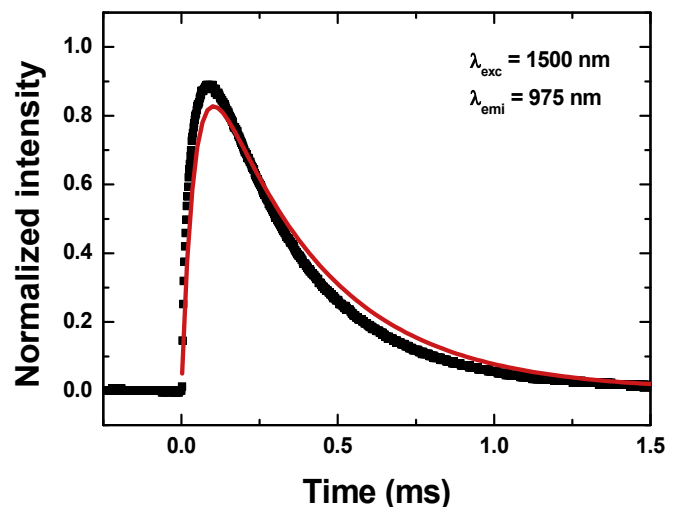


Fig. 5. Temporal evolution of $^4\text{I}_{11/2}$ level of 3.0 mol% Er_2O_3 -doped glass under pulsed laser excitation at 1500 nm. The solid red line corresponds to the theoretical fit to the experimental data using Eq. (4). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The temporal evolution of the upconversion emission after pulsed excitation at $t = 0$ can be obtained by solving Eqs. (1) and (2). The solution is given by

$$N_3(t) = \frac{W_{up}N_2(0)^2}{\frac{1}{\tau_3} - \frac{2}{\tau_2}} [\exp(-2t/\tau_2) - \exp(-t/\tau_3)] \quad (4)$$

where $N_2(0)$ is the initial population of excited ions in the $^4I_{13/2}$ level. It is remarkable in Eq. (4) that the shape of the transient $N_3(t)$ corresponds to a curve with a rise time and a decay time which is the signature of ETU, as it is obtained experimentally as shown in Fig. 5. Moreover, the temporal profile of the $^4I_{11/2}$ level for the 3.0 mol% Er_2O_3 -doped glass is shown in Fig. 5. This temporal profile has been fitted to Eq. (4) by substituting the experimental lifetimes, $\tau_2 = 0.72$ ms (for the $^4I_{13/2}$ level) and $\tau_3 = 42$ μs (for the $^4I_{11/2}$ level) and the resultant curve (solid red line) is shown in Fig. 5. As can be seen, a reasonable agreement is evident between experimental and theoretical curves, which confirms the observed UC emission due to ETU process.

3.3. Solar cell performance

From the UC studies, it is found that 3.0 mol% Er_2O_3 -doped glass exhibits an intense UC emission at 975 nm compared to 0.01 and 0.5 mol% Er_2O_3 -doped glasses (See Fig. 2). Hence, the solar cell performance has been studied for 3.0 mol% Er_2O_3 -doped glass (with a thickness of 2.5 mm). This glass sample is placed over a Si solar cell and excited with 1500 nm laser. A bright UC emission is observed when the sample excited with a laser power of 80 mW, as shown in the inset of Fig. 7. It is observed that the photocurrent increases significantly with the increase of laser pump power, a similar way of dependence that is observed in Fig. 4. The photocurrent enhances with the increase of the pump power, which may be due to the production of electron-hole pairs in the Si solar cell by the absorption of 975 nm emission. When the glass sample is removed from the solar cell no photocurrent is detected as the measurements were performed in dark. As it is expected that the laser excitation at 1500 nm itself does not produce any electron-hole pairs in the solar cell, if the glass sample is removed. Part of the surface of the solar cell is covered by the 3.0 mol% Er_2O_3 -doped glass sample (as shown in the inset of Fig. 7) and it occupies the area around 1×1.5 cm^2 .

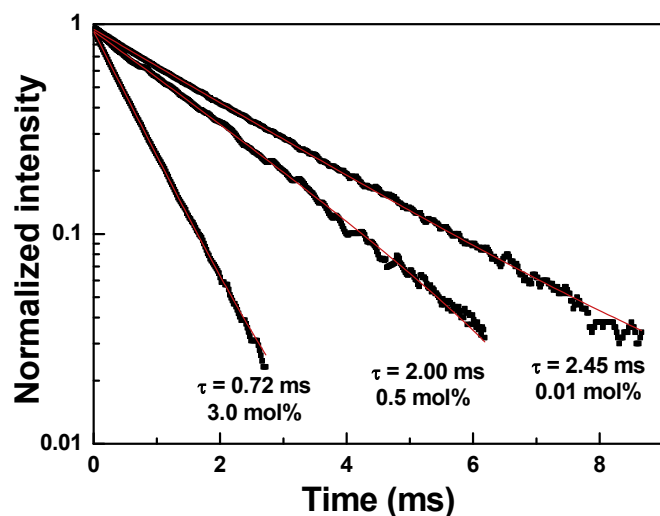


Fig. 6. Decay curves for the $^4I_{13/2}$ level of Er^{3+} ions in tellurite glasses obtained under 1500 nm excitation.

A dependence of photocurrent with the Er^{3+} concentration has also been studied for 0.01 and 0.5 mol% Er_2O_3 -doped glasses under 1500 nm laser excitation. It is observed that photocurrent has been enhanced significantly with increase of Er^{3+} concentration, as shown in Fig. 8. Taking into account that the UC emission at 975 nm depends on the square of the Er^{3+} concentration (see Eq. (3)), and expected to be a quadratic behavior. However, the experimental results do not show this quadratic nature, as shown in Fig. 8. In the previous sections, it was explained that the lifetimes of the $^4I_{11/2}$ and $^4I_{13/2}$ levels, τ_3 and τ_2 in the Eq. (3), respectively, depending on Er^{3+} concentration. Therefore, the dependence of the UC intensity or of the photocurrent are not quadratic with the Er^{3+} concentration, due to the decrease of lifetimes of excited levels when the Er^{3+} concentration is increased.

External quantum efficiency (EQE) is widely used to characterize the effect of upconverting materials on efficiency of the solar cell. EQE is defined as the ratio between the number of generated electron-hole pairs due to the UC processes and the number of incident infrared photons. The EQE is found to be 0.1% for the 3.0 mol% Er_2O_3 -doped tellurite glass excited at 80 mW power of 1500 nm laser. The EQE of 0.4% has been reported for $\text{Er}^{3+}/\text{Yb}^{3+}$ codoped fluorindate glasses upon laser excitation at 1480 nm with a laser power of 37 mW [5] and 0.27% for $\text{Er}^{3+}/\text{Yb}^{3+}$ codoped oxyhalide tellurite glasses upon laser excitation at 980 nm with a laser power of 300 mW [6]. The experimental setup used for the investigation of Er^{3+} -doped tellurite glasses is similar to the setup employed in Ref. [5]. Although the value of EQE is lower than the value obtained for fluorindate glasses, it can be enhanced by codoping with Yb^{3+} ions, synthesis conditions and optimizing the concentration of doping ions.

4. Conclusions

Photon upconversion processes in Er^{3+} -doped tellurite glasses have been investigated by exciting the $^4I_{15/2} \rightarrow ^4I_{13/2}$ transition of Er^{3+} ions with a laser at 1500 nm. The upconversion emissions centered at 548 nm, 671 nm, 800 nm and 975 nm have been observed due to energy transfer processes, evidenced by simple rate equation model. The upconversion emission at 975 nm has been enhanced enormously with the increase of Er^{3+} concentration

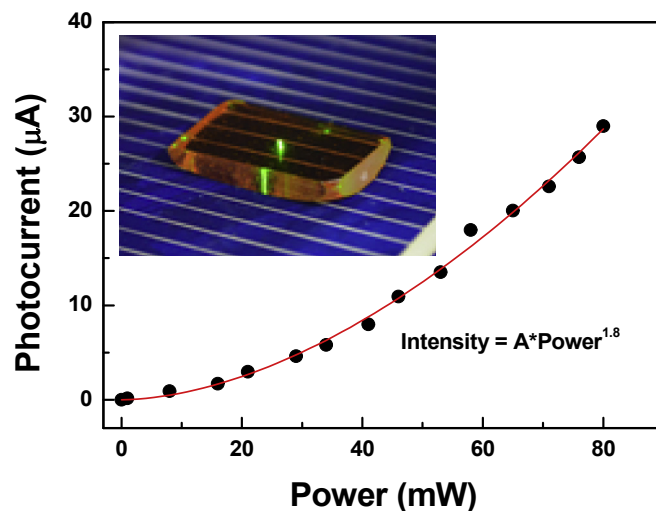


Fig. 7. Solar cell's photocurrent as a function of laser power with a laser at 1500 nm. The data points obtained for the 3.0 mol% Er_2O_3 -doped tellurite glass. Inset shows a photograph of Er^{3+} -doped glass which placed over a solar cell. The glass sample was excited perpendicularly with a laser at 1500 nm and laser power of 80 mW.

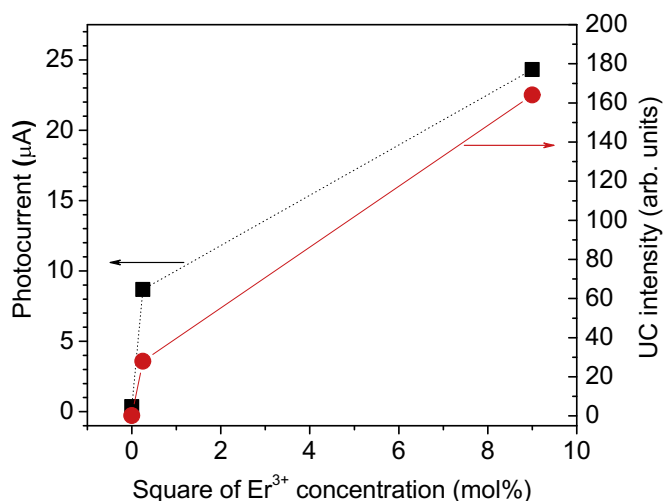


Fig. 8. Photocurrent response of solar cell with upconversion sample upon laser excitation at 80 mW of 1500 nm for different Er^{3+} concentration.

as well as pump power. Among all other emission bands, a promising emission at 975 nm (1.2 eV) of Er^{3+} ion, which is just above the band gap energy of silicon solar cells (1.1 eV), has been observed. The photons of 975 nm has been absorbed by the solar cell, enhances the photocurrent significantly upon increasing the laser pump power and Er^{3+} concentration. The studies reveal that the solar cell efficiency can be improved with Er^{3+} -doped tellurite glass through ETU process. Future work will focus on the synthesis of glasses under controlled atmosphere to enhance further the photoluminescence quantum efficiency, which may contribute in improving the solar cell efficiency.

Acknowledgments

Dr. Krishnaiah is thankful to Department of Science and Technology (DST) for the sanction of the research project (EMR/2017/000009). Dr. Venkatramu is grateful to DST, New Delhi for the sanction of international collaborative research project (No. INT/PORTUGAL/P-04/2017) under India-Portugal bilateral scientific

and technological cooperation. This work has been also supported by MINECO and EU-FEDER through the projects MAT2013-46649-C4-4-P, MAT2015-71070-REDC, and MAT2016-75586-C4-4-P. Dr. Ch. Basavapoornima is also thankful to University Grants Commission, New Delhi, for the award of Women Post-Doctoral Fellowship during the year, 2011-12 (F.15-1/2011-12/PDFWM-2011-12-OB-AND-9964 (SA-II), dt. 1-11-2013).

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